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Introduction to Robotics (Spring 2026)

Design and Implementation of an Ultrasonic-Based Pick-and-Place Mobile Manipulator Robot

Configuration: RRR Manipulator on a Mobile Base

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1 Introduction & Applications

1.1 Introduction

In modern industrial automation, integrating mobile platforms with robotic manipulators creates highly flexible and intelligent systems. This report presents the design and implementation of a Bluetooth-Controlled Mobile Platform with an Autonomous Ultrasonic-Based Pick-and-Place Manipulator.

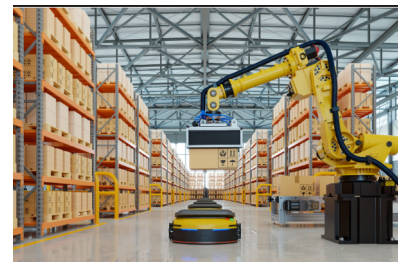
The system is split into two operating modes to balance human guidance with automation:

- **The Mobile Base (Manual/Bluetooth Mode):** A two-wheeled differential drive chassis governed by an ESP32 microcontroller. It receives wireless steering commands from a custom mobile application via Bluetooth Classic, allowing an operator to drive the robot precisely toward target zones.
- **The Robotic Manipulator (Autonomous Mode):** A 3-Degree of Freedom (DoF) Articulated Robotic Arm controlled by an Arduino Uno. To achieve precise autonomous grasping, an Ultrasonic distance sensor is mounted on the system. Once the operator positions the vehicle near an object, the ultrasonic sensor dynamically measures the exact distance to the target. Based on this real-time feedback, the system calculates and adjusts the joint angles of the arm to accurately reach, grab, and place the object in a pre-programmed sequence.

1.2 Applications of Wheeled Mobile Sorting Manipulators

The integration of a 5-DoF articulated arm with a mobile platform creates a highly versatile robot capable of operating in dynamic environments. Key real-world applications include:

Smart Warehouse Logistics: Autonomous mobile manipulators can navigate through warehouse aisles, detect inventory packages using proximity sensors, and transport items between picking zones without human intervention.



Automated Material Handling (AMH): In flexible manufacturing lines, these robots serve as a link between different stationary machines, picking up fabricated parts from one assembly station and placing them onto another.



Hazardous Operations & EOD (Explosive Ordnance Disposal): Mobile manipulators can be deployed in environments that pose severe risks to human safety, such as chemical plants, nuclear facilities, or bomb disposal zones, to pick up and isolate suspicious objects from a safe distance.



E-Commerce Order Fulfillment: With the exponential growth of e-commerce, these systems automate the sorting, consolidating, and retrieving of multi-item orders in sorting facilities, drastically reducing processing time.



2 List of Electrical and Mechanical Components

The following tables outline the comprehensive bill of materials utilized to construct both the mobile platform and the 3-DoF robotic manipulator.

2.1 Mechanical Components

Component Name	Description / Specification	Qty	Component Image
5-DoF Robotic Arm Chassis Kit	3D printing structural links that form the articulated arm body, including the end-effector.	1	
2WD Smart Robot Car Chassis Kit	Acrylic base plate template designed for differential drive locomotion. Includes motor supports.	1	
Mounting Brackets & Fasteners	Screws (M3/M4), nuts, standoffs, and servo horns required for secure assembly.	1 Set	

Table 1: Mechanical Components List

2.1.1 Arm geometry

Joint / Link	Length (cm)
Joint 1 / Link 1	4.8
Joint 2 / Link 2	11.2
Joint 3 / Link 3	9.4
Joint 4 / Link 4	2.8
Joint 5 / Link 5	10.8
Joint 6 / Gripper	—

Table 2: Geometric Specifications and Lengths of Robot Joints and Links

2.2 Electrical & Electronic Components

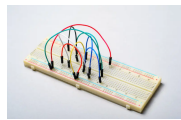
Component Name	Description / Specification	Qty	Component Image
Arduino Uno R3	Microcontroller board dedicated to handling the arm movements and kinematics.	1	
ESP32 Microcontroller	High-performance controller dedicated to managing the mobile base navigation.	1	
MG995 Servo Motor	High-torque metal gear servo motors positioned at the lower heavy-load joints.	2	
SG90 Micro Servo Motor	Lightweight 9g nylon-gear servo motors positioned at upper lightweight joints.	3	
PCA9685 Servo Driver	16-Channel, 12-bit PWM I2C Servo Driver to prevent signal jitter.	1	
L298N Dual Driver	High-power motor driver module used to control DC gear motors via PWM.	1	
BO DC Gear Motors	Yellow plastic gear motors (3V-6V) providing locomotion torque.	2	
Ultrasonic Sensor	Proximity and distance sensor used to detect the presence and measure the distance of an upcoming object via ultrasonic sound waves.	1	
Power Supply / Batteries	Li-ion batteries arranged to provide separate adequate voltage channels.	1 Set	
Breadboard & Wires	Prototyping board along with jumper wires for circuit connections.	1 Set	

Table 3: Electrical and Electronic Components List

3 Schematic and Electrical Interconnections

The electrical architecture is carefully designed to handle wireless communication on the mobile base and real-time sensor processing on the manipulator arm.

3.1 Subsystems Breakdown

1. **Mobile Base Subsystem (ESP32):** The ESP32 utilizes its internal Bluetooth module to establish a wireless link with a smartphone controller app. Based on the received control packets, the ESP32 drives the L298N Dual H-Bridge module via digital PWM signals to control the locomotion of the two BO DC gear motors.

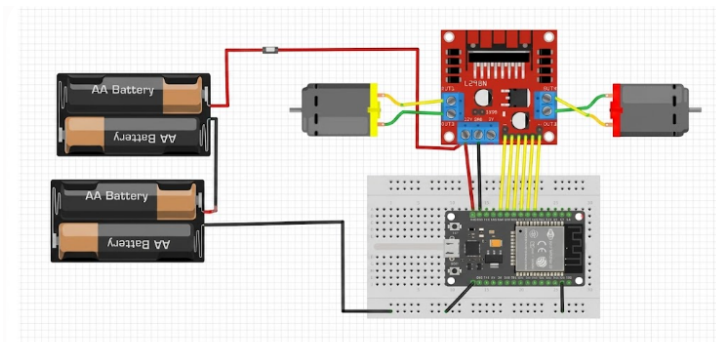


Figure 1: Mobile Base Subsystem Schematic Diagram.

2. **Robotic Arm Subsystem (Arduino Uno):** The Arduino Uno runs the automation loop for the 3-DoF manipulator. An Ultrasonic sensor (HC-SR04), mounted directly on the system, is used to measure the exact distance to the target object. The sensor is wired to the Arduino via Trigger and Echo pins, where it sends high-frequency sound waves and measures the time taken for the echo to return.

Once the object is within the detection range, the Arduino processes the time-of-flight data to calculate the real-time distance. Based on this precise measurement, the microcontroller dynamically computes the required joint angles for the 3-DoF arm to align the end-effector with the target. The Arduino then commands the PCA9685 16-Channel Servo Driver over the I2C bus (SDA/SCL) to actuate the 5 servo motors ($2 \times$ MG995 and $3 \times$ SG90) smoothly, executing the automated pick-and-place sequence.

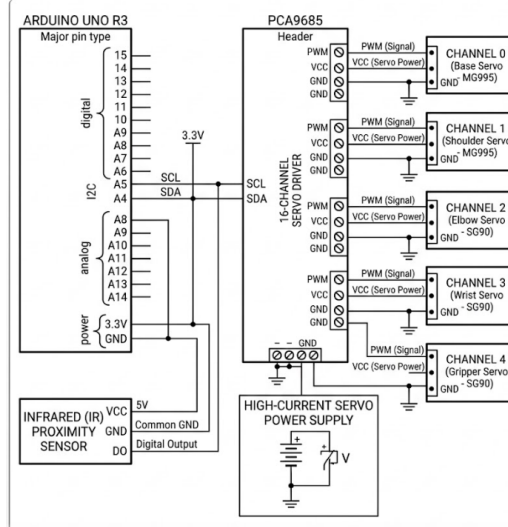


Figure 2: Robotic Arm Subsystem Schematic Diagram.

3.2 Power Management and Ground Isolation

Due to the significant current spikes drawn by the high-torque MG995 servo motors and the DC gear motors under load, a robust power distribution network is implemented:

- **Logic Power:** The microcontrollers (Arduino Uno and ESP32) are powered through stable, regulated 5V/3.3V rails to prevent resets caused by voltage drops (brownouts) during motor transients.
- **Actuator Power:** The PCA9685 servo driver is connected to an external high-current power source (e.g., Li-ion battery configuration supplying 6V–7.4V) dedicated purely to powering the servo motors. The L298N motor driver is powered independently via a separate higher voltage rail to satisfy the DC motors' requirements.
- **Common Ground:** Crucially, the grounds (GND) of the Arduino Uno, ESP32, PCA9685 driver, L298N driver, and all battery packs are tied together to create a common reference plane, ensuring reliable logic signals and preventing floating voltage faults.

4 Forward and inverse Kinematics using Simulink

4.1 System block diagram

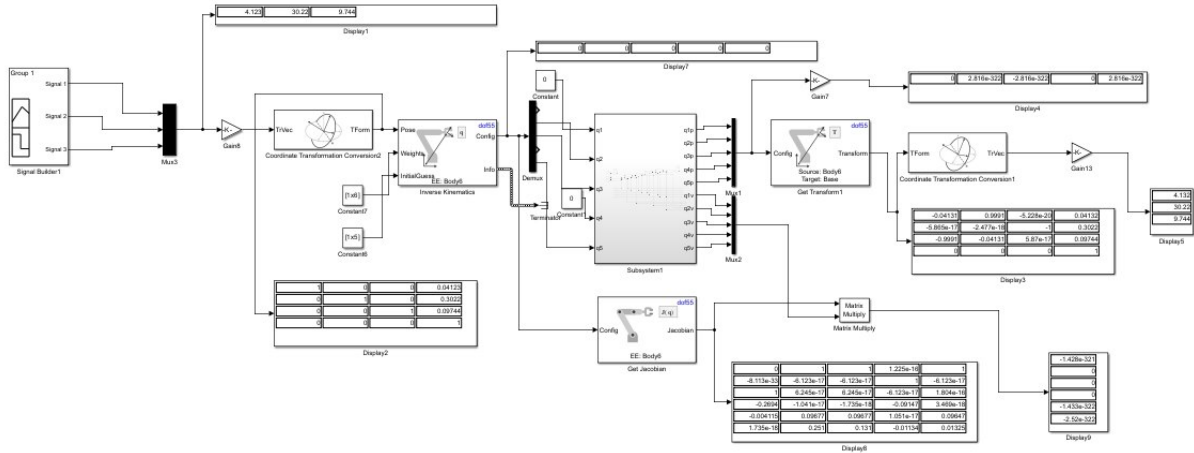
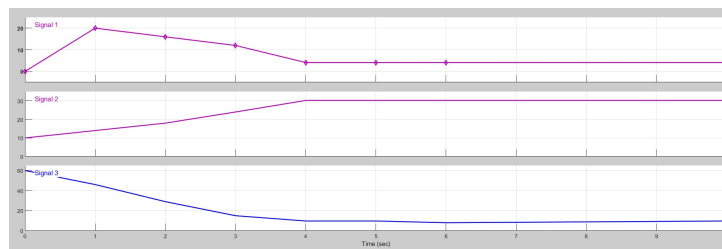


Figure 3: System block diagram.

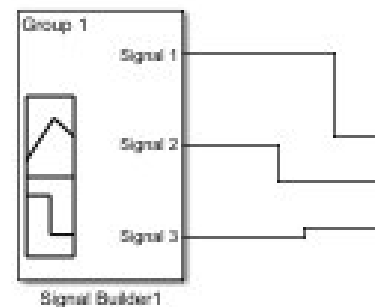
This system is designed for the kinematic and differential analysis of a 3-Degree-of-Freedom (3-DOF) robotic manipulator. It takes a desired trajectory, converts it into joint space using Inverse Kinematics, passes it through a robot subsystem, and verifies the results using Forward Kinematics (Transform) and a Jacobian matrix.

4.2 System block diagram explanation

4.2.1 Signal Builder

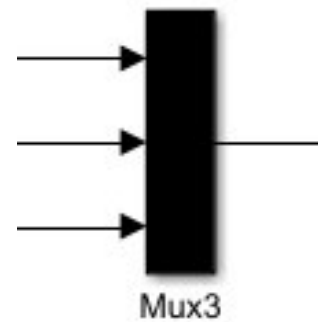


- **Explanation:** This block generates custom, time-varying input signals. In this context, it acts as a trajectory generator providing the desired X, Y, Z Cartesian coordinates for the robot's end-effector over time.
- **Input:** None (Internal time-based configuration).
- **Output:** Three separate scalar signals (Signal 1, Signal 2, Signal 3) representing the desired position vector.



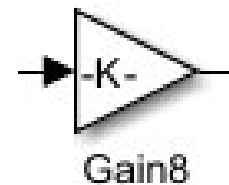
4.2.2 Mux3 (Multiplexer)

- **Explanation:** Combines multiple scalar signals into a single vector signal to keep the routing clean and compatible with vector-input blocks.
- **Input:** 3 separate scalar signals from Signal Builder.
- **Output:** A single 3-element vector $[X, Y, Z]^T$.

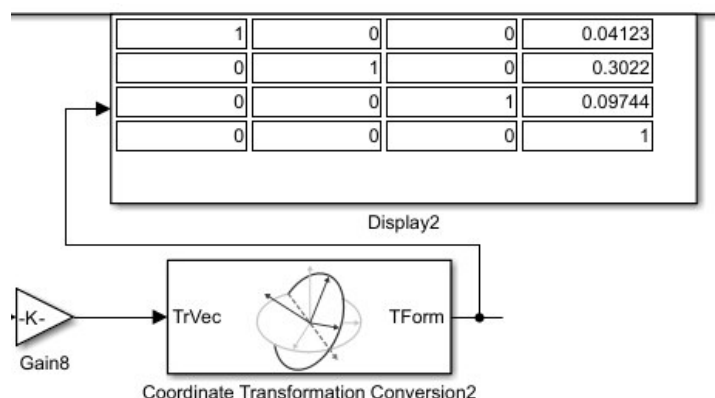


4.2.3 Gain

- **Explanation:** Simple multiplication blocks. They are likely used here as scaling factors (e.g., converting meters to millimeters, or adjusting coordinate scaling).
- **Input:** A vector or scalar signal.
- **Output:** The input signal multiplied by the constant gain factor (K).

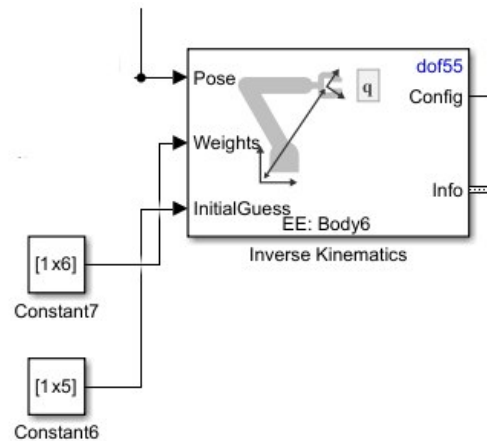


4.2.4 Coordinate Transformation Conversion (TrVec $\rightarrow$ TForm)



- **Explanation:** Converts a 3-element translation vector $[X, Y, Z]$ into a 4×4 homogeneous transformation matrix. Because no orientation is provided, it assumes a default identity rotation matrix.
- **Input:** A 3-element translation vector (seen in Display1 as roughly $[4.123, 30.22, 9.744]$).
- **Output:** A 4×4 homogeneous transformation matrix (displayed in Display2).

4.2.5 Inverse Kinematics

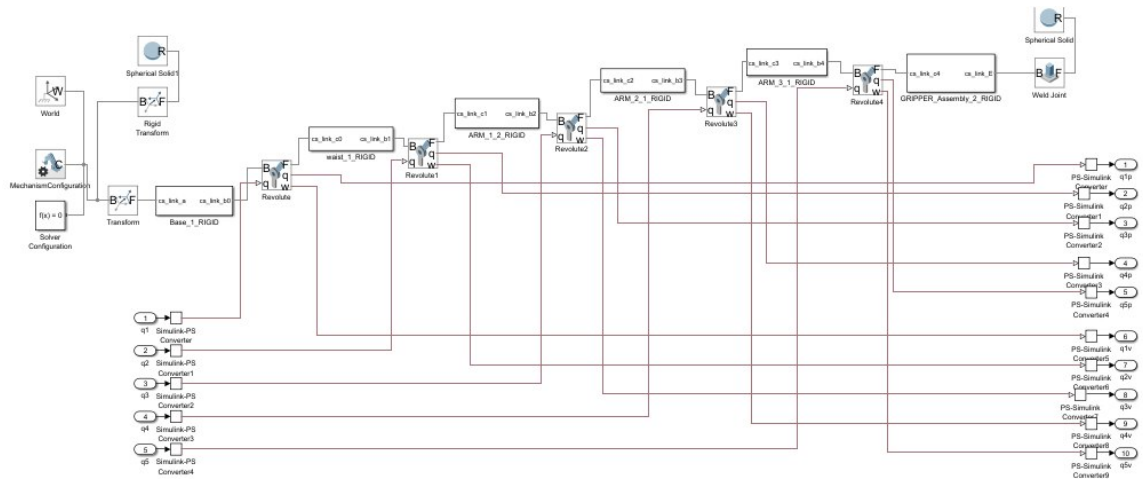


- **Explanation:** This is a core Simscape Multibody block. Given a desired end-effector pose (position and orientation), it calculates the required joint angles (q) for a 5-DOF manipulator (dof55) to reach that target.
- **Input:**
 - Pose: Desired 4×4 transformation matrix.
 - Weights: Importance weighting for position vs. orientation tracking (from Constant7).
 - InitialGuess: Starting joint configurations to prime the numerical solver (from Constant6).
- **Output:** Config: A vector of the calculated 5 joint angles.

4.2.6 Demux (Demultiplexer)

- **Explanation:** Splits a combined vector signal back into individual scalar components.
- **Input:** The 5-element joint configuration vector from the Inverse Kinematics block.
- **Output:** 5 individual scalar lines, each representing a specific joint angle (q_1 to q_5).

4.2.7 Subsystem



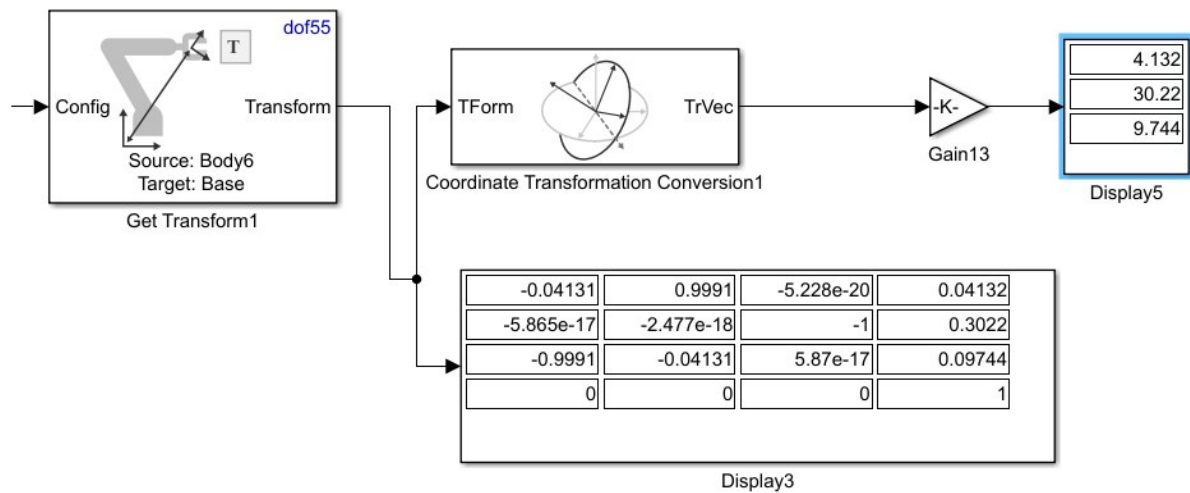
- Explanation:** This block represents the actual physical or virtual plant model of the 3-DOF robot manipulator. It takes the target joint positions, drives the joints, and simulates the physical movement to output the actual positions (q) and velocities (qv).
- Input:** 5 individual joint position commands (q_1 through q_5).
- Output:**
 - 5 actual joint positions ($q1p$ to $q5p$).
 - 5 actual joint velocities ($q1v$ to $q5v$).

4.2.8 Get Transform (Forward Kinematics)

- **Explanation:** performs Forward Kinematics. It reads the current joint configurations (q) and computes the resulting coordinate transformation matrix from the base of the robot (Target: Base) to the end-effector (Source: Body6).
- **Input:** Config (The 5-element joint position vector from Mux1).
- **Output:** 4×4 homogeneous transformation matrix showing the actual current pose of the robot (displayed in Display3).

4.2.9 Coordinate Transformation Conversion1 (TForm $\rightarrow$ TrVec)

- **Explanation:** The inverse of block 4. It extracts the 3-element translation vector $[X, Y, Z]$ out of a 4×4 homogeneous transformation matrix.
- **Input:** A 4×4 transformation matrix from Get Transform1.
- **Output:** A 3-element position vector (sent to Gain13 and shown on Display5 to verify it matches the original input in Display1).

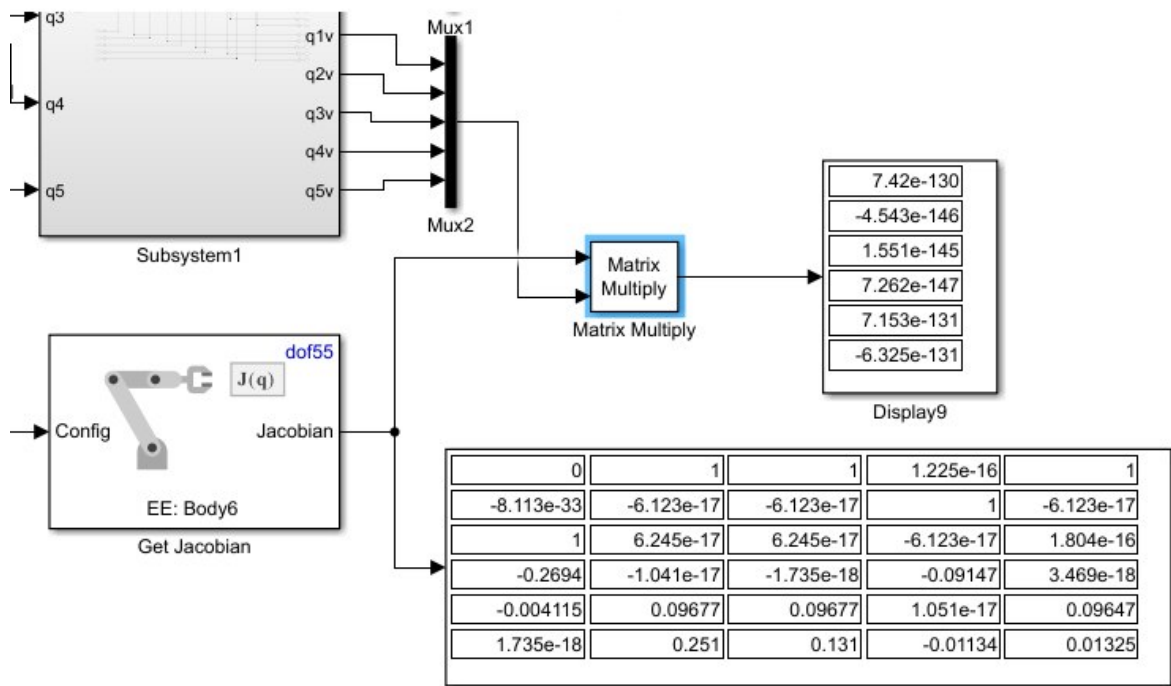


4.2.10 Get Jacobian

- **Explanation:** Computes the geometric Jacobian matrix (J) for the current robot configuration. The Jacobian maps joint velocities to Cartesian linear and angular velocities ($\dot{x} = J(q)\dot{q}$).
- **Input:** Config (The 5-element joint position vector).
- **Output:** A 6×5 Jacobian matrix (displayed in Display8).

4.2.11 Matrix Multiply

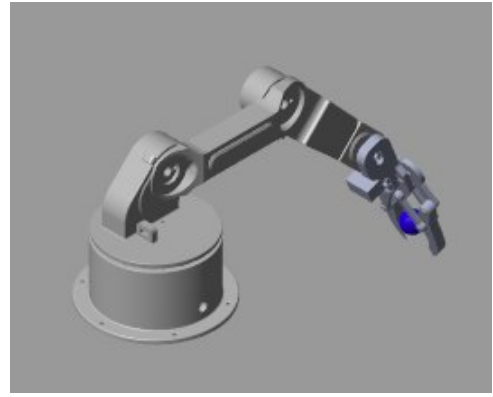
- **Explanation:** Performs standard algebraic matrix multiplication. Here, it multiplies the 6×5 Jacobian matrix by the 5×1 joint velocity vector ($J \cdot \dot{q}$).
- **Input:**
 - the 6×5 Jacobian matrix.
 - The 5×1 joint velocity vector from Mux2.
- **Output:** A 6×1 Cartesian velocity vector (3 linear velocities, 3 angular velocities) of the end-effector, displayed in Display9.



4.3 Results

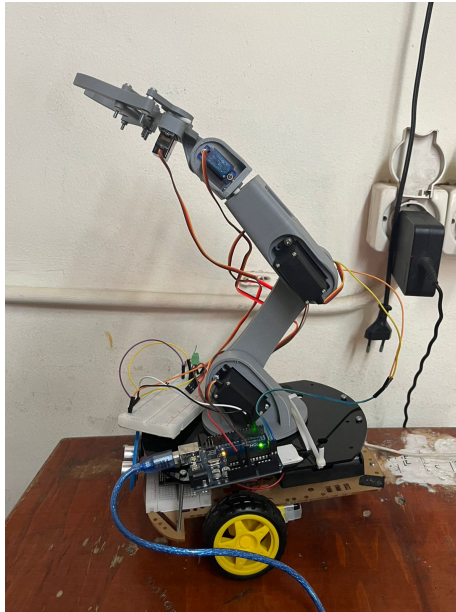


(a) Initial position



(b) Final position

Figure 4: Initial and final configurations of the robot's end-effector during the pick-and-place simulation.



(a) Initial position

5 Budget of the project

Component Name	Price
Arduino Uno	400
ESP32	400
2*MG995 Servo Motor	340
3*SG90 Micro Servo	225
PCA9685 Servo Driver	190
L298N Dual Driver	75
Ultrasonic Sensor	40
Power Supply / Batteries	80
2WD Smart Robot Car Chassis Kit	300
3D Robot Mechanical Arm	650
Total Cost	2700

Table 4: Budget

6 Challenges and Solutions

During the design and implementation phases of the mobile manipulator, several technical challenges were encountered. These obstacles were systematically addressed to ensure the system's structural integrity and operational efficiency.

1. Mechanical Structure and Material Durability

- **Challenge:** The initial prototype utilizing 3D-printed parts suffered from poor material quality, leading to structural fragility. Additionally, the selected motors lacked the necessary torque to handle and support the weight of the 5-DoF manipulator and the mobile base effectively.
- **Solution:** To overcome these mechanical limitations, the entire physical design was optimized, and the dimensions were re-engineered. Furthermore, the low-torque motors were replaced with high-performance, higher-torque variants capable of seamlessly managing the required payload and dynamic movements.

2. Power Supply and Management

- **Challenge:** Initially, standard 1.5V batteries were used to power the system. However, this configuration provided insufficient power, failing to maintain stable Bluetooth communication while simultaneously driving the motor driver.
- **Solution:** The power subsystem was upgraded by integrating 3.7V Lithium-Ion rechargeable batteries. This upgrade significantly improved power efficiency, providing a stable voltage source capable of running the ESP32 microcontroller, sustaining continuous Bluetooth connectivity, and delivering ample current to the motor driver.

7 References

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